

Negative indices



1 Consider the following expressions:

i Identify the base.

ii Identify the power.

a 10^{-7}

b 2^{-4}

c 13^{-10}

d $(-5)^{-8}$

2 Complete the following tables:

a

2^5	2^4	2^3	2^2	2^1	2^0	2^{-1}
32	16					

b

10^5	10^4	10^3	10^2	10^1	10^0	10^{-1}
100 000	10 000					

c

3^3	3^2	3^1	3^0	3^{-1}	3^{-2}	3^{-3}
27	9					

3 Express the following expressions with a positive index:

a 6^{-10}

b 73^{-14}

c $(-9)^{-7}$

d 9^{-1}

e 17^{-6}

f 55^{-1}

g $(-12)^{-8}$

h -45^{-5}

i -8^{-11}

j $(-20)^{-3}$

k 7^{-6}

l $(-5)^{-1}$

4 Express the following expressions with a negative index:

a $\frac{1}{3}$

b $\frac{1}{37}$

c $\frac{1}{5}$

d $\frac{1}{4^7}$

e $\frac{1}{-15^3}$

f $\frac{1}{10^5}$

g $\frac{1}{(-24)^{10}}$

h $\frac{1}{25^3}$

i $\frac{1}{13^{11}}$

j $\frac{1}{7^8}$

k $\frac{1}{16^{12}}$

l $\frac{1}{(-45)^7}$

5 Complete the following statements:

a $\frac{1}{3^2} = 3^{\square}$

b $\frac{1}{6^5} = 6^{\square}$

c $\frac{1}{27} = 3^{\square}$

d $\frac{1}{5^3} = 5^{\square}$



6 Simplify the following expressions:

a $5^{11} \div 5^{-3}$

b $7^{-7} \div 7^5$

c $7^{-3} \times 7^{-4}$

d $5^{-4} \div 5^{-9}$

e $9^0 \times 9^{-12}$

f $5^{13} \div 5^{-9}$

g $10^2 \div 10^3$

h $100^{25} \div 100^{26}$

i $\frac{2^3}{2^5}$

j $\frac{(5^2)^9 \times 5^6}{5^{40}}$

k $\frac{(19^2)^3}{19^{-3} \times 19^{-9}}$

l $\frac{-(8^5)^2}{8^{-2} \times 8^{-12}}$

7 Evaluate:

a $4 \times 3^{-2} + 8^0$

b $6 \div 2^{-1} + 4$

c $(2 + 3^{-2}) \times 2^{-1}$

d $\left(\frac{2}{3}\right)^{-3} \div 6^{-1}$

8 Answer the following questions:

a What is 0^4 equal to?

b Explain why 0^{-4} is undefined.

9 Evaluate the following expressions:

a $6^7 \times 6^{-7}$

b $4^8 \times 4^{-6}$

c $3^{-8} \times 3^{11}$

d $5^5 \times 5^{-7}$

e $4^{-4} \times 2^{-4}$

f $12^{13} \div 12^7 \div 12^8$

g $5^{-2} \times 3^{-2}$

h $(-4)^3 \times (-4)^{-7}$

i $7^{12} \div 7^{-7} \div 7^{17}$

j $\frac{3^{-9} \times 3^{-7}}{3^{-14}}$

k $\frac{5^7 \times 5^{-8}}{5^{-4}}$

l $\frac{(-7)^{-11} \times (-7)^{-4}}{(-7)^{-14}}$